

Exercise 4

- ★ Suppose the excited nucleus B^* recoils with speed $v \ll c$ and emits a photon of energy E_γ in its rest frame.

- Show that in the laboratory frame, to first order in v/c

$$E_\gamma^{\text{lab}} \simeq E_\gamma \left(1 + \frac{v}{c} \cos \theta \right),$$

where θ is the emission angle with respect to the recoil direction.

- Which emission direction maximizes the photon energy?
- Explain why resonant absorption preferentially selects photons emitted approximately along the recoil direction of B^* .

